

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)		2 materials chosen and microscopic structure of each explained [2 × (1)] See below. 1 example given of each material [2 × (1)] See below. Crystalline - long range order / lattice like arrangement/regular arrangement structure e.g. metals Amorphous - short range order / irregular or random arrangement / no order e.g. glass, ceramics, brick Polymeric - long chain molecule arrangement [of hydrocarbons] e.g. rubber, polythene accept plastic	4			4		
	(b)		Indicative content: Measurements: - Extension of wire [with pointer/ruler] - Original length of wire [from clamp to pointer] - Diameter of wire using micrometer or (Vernier) callipers Determination of Young modulus: - Use $E = \frac{Fl}{Ae}$ or plot graph of load/extension or stress/strain and find gradient A determined from $\pi \left(\frac{d}{2}\right)^2$ Precautions: - Repeat readings of e (adding/removing load) - Measure $\{d$ in various places on wire / mean $d\}$ - Keep temperature constant / use of Searle's apparatus - Ensure no kinks in wire - Soft wood so as not to damage wire - Ensure wire is securely clamped - Stay within elastic limit - Use of a longer wire / travelling microscope - Avoid parallax	4		2	6		6